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PRINCIPLES OF GAUGE THEORY OF REAL HADRONS

1. Non-Yang-Mills gauge theories.

The basic assumptions about hadron theory

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Дана физическая интерпретация и математическая формулировка понятия калибровочного преобразования как порождаемого любым представлением данной группы. Высказано предположение, что теория адронов инвариантна относительно калибровочных преобразований, порождаемых векторным представлением полной группы Лоренца.

The paper "Principles of gauge theory of real hadrons" consists of three sections:

1. Non-Yang-Mills gauge theories. The basic assumptions about hadron theory.

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2. Lorentz-vector gauge invariance

(Preprint INP MSU-92-13/262)

3. Lorentz-spinor gauge invariance

(Preprint INP MSU-92-14/263)

4. Interaction of hadrons in Lorentz-vector gauge theory

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Abstract to four sections of the paper.

The studies that resulted in this work were based on the conviction that inefficiency of the present theory of hadrons should not be overcome without abandoning the description of nucleons by a Dirac spinor. Thereby the aim of these studies was to find a mechanism which necessarily selects some representations of the Lorentz group from the continuum of available representations containing the spin $1/2$. The work contains three basic points. First, we give a new physical interpretation and a new mathematical formulation of gauge transformations. These transformations are generated by any representation of the given group. In the framework of such formulation the Yang-Mills gauge transformations are a partial case: these are gauge transformations generated by the adjoint representation of a Lie group. Second, we formulate the following assumption: the hadron theory is invariant under gauge transformations generated by the vector representation of the full Lorentz group. We perform a complete analysis of relativistically-invariant equations corresponding to such theories. A field that satisfies some of these equations cannot be finite-dimensional. Each of admissible representations of the Lorentz group associated with such a field can be decomposed into a direct infinite sum of finite-dimensional irreducible representations. Third, it is supposed that the hadrons acquire masses in accordance with the following scheme consisting of two parts: 1) there exist Higgs bosons that interact with hadrons and are described by trivial Lorentz-vector gauge transformations (an operator of each transformation coincides with the unit operator); 2) Lorentz-scalar components of neutral meson fields have non-zero vacuum expectation value, i.e. the mesons can be regarded as Higgs bosons described by non-trivial Lorentz-vector gauge transformations. The trivial Higgs bosons themselves would lead to continuous infinitely degenerated, in respect to spin, mass spectrum of hadrons. We find infinitely-dimensional matrix operators whose elements give coupling of different field components in the interaction Lagrangian which is invariant under gauge transformation. As one of qualitative consequences of the theory we present a new point of view to the problem of the decay $K_L^0 \rightarrow \pi^+ \pi^-$. According to it the π -mesons are not pure states with the spatial parity $P = -1$; rather, they have an admixture with $P = +1$.

1 Non-Yang-Mills gauge theories. The basic assumptions about the hadron theory

1.1 Introduction

Hardly somebody can argue against the fact that theoretical constructions in the theory of hadrons lead to much worse numerical results than the theory of leptons, or they are not able to give any results. Numerous experiments with attempts to discover fractionally charged particles have been failed. For almost twenty last years one could not invent any rigorous proof of the idea of bags for quarks and gluons. To a large extent, the new atomism uses phenomenological approaches both in solving the problem about the spectra of masses of baryons and mesons and in calculating various characteristics of interaction processes with hadrons. We can mention the spin crisis [1] as a striking example of unreliability of our concepts about nucleons.

Usually one believes that the level of descriptive and predictive power of the existing theory of hadrons is connected with the nature of the strong interaction itself, rather than with the concepts that form the base of this theory.

I do not argue against any convictions. Rather, I believe in construction of a theory of hadrons that will not involve unobservable particles and will describe equally well all the properties and interactions of hadrons at all energies. The future theory should inevitably rest on new initial principles. When searching such principles it is instructive to exploit the whole existing variety of theoretical ideas and experimental results aiming to find unrealized possibilities in analyses and generalizations.

What serves as a source of present representations about nucleons? In the first place, this is the interpretation of Hofstadter's experiment [2] on elastic scattering of electrons on protons. The experimental results have been compared with the Rosenbluth's formula [3] for the cross-section of elastic electromagnetic interaction of the electron and a particle which is described by a Dirac spinor (bispinor of the first rank of the proper Lorentz group), on the one hand and has an arbitrary magnetic moment and finite sizes on the other hand. The data obtained admitted themselves the following two interpretations: 1) in contrast to the electron, the nucleon has its own structure; 2) the nucleon does not have a structure, but, in contrast to the electron, it cannot be described by a Dirac spinor. It was the first interpretation of the experiment that was predetermined by all the concepts connected with the nature of the nucleon and formed up to that time.

Then this idea about the structure of the nucleon found its concrete realization in the quark model [4]. Note that quarks are also described by a Dirac spinor. At present, in a set of structure elements of the nucleon (and other hadrons) one also includes gluons which are gauge bosons associated with the group of internal symmetry SU(3). These are internal symmetries, with a great variety of groups applied to describe them that dominate in modern theoretical studies in particle physics.

The second interpretation of the Hofstadter's experiments did not have notable studies that would aim to solve the following problem: Which representation of the Lorentz group is nucleon described by, and which equation does its wave function obeys to?

It is well known that there is a countable set of finite-dimensional and continuum of infinite-dimensional irreducible representations of the Lorentz group which contain spin $1/2$. Depending on a choice of the representation, we possess, together with the

spin $1/2$, a finite or an infinite series of other half-integer spins following one after another with the unit step. If we want to describe the nucleon in such a way, then we will inevitably get particles with higher spins ($3/2$ etc.), i.e. nucleon resonances. Generally speaking, the spin is not invariant under Lorentz transformations, so that the assertion that a particle has the spin s is associated with its rest frame.

If all the nucleon resonances are described by a single relativistically-invariant equation, then the corresponding representation of the Lorentz group is most likely infinitely-dimensional. A sufficiently general analysis of the structure of linear relativistically-invariant equations, both finite- and infinite-dimensional, has been given by Gel'fand and Yaglom [5]. In [6], one can find some clarifications of conclusions concerning the dependence of mass spectra on the spin for equations constructed on a finite sum of infinite-dimensional irreducible representations of the Lorentz group. The relativistically-invariant equations constructed on an infinite sum of finite-dimensional irreducible representations of the Lorentz group contain a countable set of arbitrary constants. This is the reason why one cannot find in literature any analysis of consequences following from equations of such type.

To solve the above problem about the equation that describes the nucleon it is necessary to use some extraordinary and, in some respects, very simple approach. Let us turn to a great example of theoretical investigations of physicists by following the development of the idea of Yang-Mills gauge invariance [7] which eventually lead us to the construction of the theory of electroweak interaction [8, 9] whose basic consequences have been perfectly confirmed by experiments.

Yang and Mills proposed a treatment of gauge transformations as group transformations whose parameters depend on a point of the space-time. This treatment excites a general admiration. However, if we give it up we will be able to find a new point of view on the gauge transformations.

Let us now turn to the quantum mechanics in order to sketch the pre-history of the Yang-Mills idea. The invariance of quantum-mechanical equations under a phase transformation of the wave function $\Psi(x) \rightarrow \exp(-i\theta)\Psi(x)$ (with θ arbitrary constant) as well as the superposition principle are two sides of the same fundamental essence of nature. The superposition principle provides the difference between quantum mechanics and classical mechanics, as well as between quantum determinism and Laplace determinism. This difference cannot be removed by introducing some hidden parameters and static distribution of quantities which characterize the state of a particle. It is the phase invariance that gives, in some explicit form, a description of the freedom that is inherent in the field of the particle. The freedom is the primary property of the field and, I believe, the fact that phase transformations form the group $U(1)$ is a secondary property.

The modern state of theory and experiment enables us to conclude that the nature admits an extension of the phase invariance to the gauge invariance: $\Psi(x) \rightarrow \exp[-i\theta(x)]\Psi(x)$, up to appropriate transformations in gauge (compensating) fields. At least both for charged and neutral leptons, the validity of such statement has been proved by the development of the theory of electroweak interaction and the discovery of Z^0 -boson.

The interpretation of the phase and the gauge invariance as the freedom inherent in the field enables us to come to a new definition of phase and gauge transformations.

1.2 New formulation of the concept of gauge transformation

Definition. Suppose that there is a group G and two its representations T and S . Let $\theta = \{\theta_a\}$ be a vector in the representation space of T , $\Psi(x)$ be a field that transforms as the representation S , and D^a be some matrix operators that act on field components $\Psi_a(x)$ in the representation space of S . Then a transformation of the field Ψ

$$\Psi(x) \rightarrow \Psi'(x) = \exp(-iD^a\theta_a)\Psi(x) \quad (1.1)$$

will be called a generalized phase transformation, if the field Ψ' transforms as the same representation S of the group G .

Thus, if an element g of the group G generates the transformations

$$\Psi'_a(x) = [S(g)]_{ab}\Psi_b(x), \quad (1.2)$$

$$\theta'_a = [T(g)]_{ab}\theta_b, \quad (1.3)$$

then, for all $g \in G$, the matrix operators D^a should satisfy the following equation:

$$D_{ab}^a = [S^{-1}(g)]_{ca}D_{cb}^a[S(g)]_{cb}[T(g)]_{ba},$$

or, in a brief form,

$$D^a = S^{-1}(g)D^a S(g)[T(g)]_{ba}. \quad (1.4)$$

If G is a Lie group, then transformations (1.2) and (1.3) can be written as

$$\Psi'(x) = S(g)\Psi(x) = \exp(-iL^a\epsilon_a)\Psi(x), \quad (1.5)$$

$$\theta' = T(g)\theta = \exp(-iM^a\epsilon_a)\theta, \quad (1.6)$$

where L^a and M^a are generators of the group G , respectively, on representations S and T , $\epsilon_a = \epsilon_a(g)$ are group parameters corresponding to the given element g .

By use of (1.5) and (1.6), condition (1.4) takes the form

$$[L^a, D^a] = D^a M^a. \quad (1.7)$$

It is obvious that generalized phase transformation (1.1) exists if, and only if among irreducible representations of the direct product $T \otimes S$ (defined over the space of vectors $\theta \otimes \Psi$) there is at least one which belongs to the representation S . It is not difficult to find a situation when a transformation of the type (1.1) cannot be called generalized phase transformation, e.g., $G = SU(2)$, T is the irreducible representation of spin 1/2, and S is the irreducible representation of spin 1.

Elements of the matrix operator D^a are in fact unnormalized Clebsch-Gordan coefficients for the corresponding irreducible representations of the group G . More precisely, this statement can be formulated as follows. Let ω_i, ω_j , and ω_k denote three irreducible representations of the group G such that $\omega_i \in T, \omega_j \in S, \omega_k \in T \otimes S$, and let a_0, α_0, β_0 be a set of indices that characterize vectors of the corresponding spaces of irreducible representations. Then we have

$$D_{\alpha_0\beta_0\mu_0\mu_0}^{a_0\omega_0} = 0, \text{ if } \omega_k \ni S, \quad (1.8)$$

$$D_{\alpha_0\beta_0\mu_0\mu_0}^{a_0\omega_0} = C_{\alpha_0\omega_0\mu_0}^{a_0\omega_0} C_{\beta_0\omega_0\mu_0}^{a_0\omega_0} \text{ if } \omega_k \in S, \quad (1.9)$$

where $C_{\alpha_1 \beta_1 \alpha_2 \beta_2 \alpha_3 \beta_3}$ is an appropriately normalized Clebsch-Gordan coefficient, $d_{\alpha_1 \beta_1 \alpha_2 \beta_2}$ is an arbitrary quantity independent of the indices $\alpha_3, \alpha_0, \beta_3$. Applying (1.9) we will often use an expression of the type: 'matrix operator D^a is determined by the quantities $d_{\alpha_1 \beta_1 \alpha_2 \beta_2}$ '.

We turn from generalized phase transformation (1.1) to the gauge ones in a clear way: the parameters θ_a should be understood as functions of the space-time coordinates, and it is necessary to appropriately introduce the gauge (compensating) fields and their transformations. In what follows, I will refer to formula (1.1) as to the gauge transformation generated by the representation T of the group G , keeping in mind all that was said in the previous sentence.

Let us prove that the class of transformations determined by our definition includes, as a partial case, Yang-Mills gauge transformations, but with another interpretation of the quantities involved.

Thus let us take a Lie group G as well as its representation S and a field $\Psi(x)$ that transforms as this representation. Then to each element $g \in G$ there corresponds transformation (1.5). Commutation relations for generators L^a have the form

$$[L^a, L^b] = if^{abc} L^c, \quad (1.10)$$

with f^{abc} structure constants of the group G .

Let us now consider a generalized phase transformation (1.1) of the same field $\Psi(x)$ and let us assume that the vector $\theta = \{\theta_a\}$ transforms as the adjoint representation of the group G , i.e. the matrix operators M^a in (1.6) are such that

$$M^a_{ab} = if^{abc}. \quad (1.11)$$

Then (1.7) and (1.11) show that the operators D^a should obey the following equation:

$$[L^a, D^b] = if^{abc} D^c. \quad (1.12)$$

By comparing (1.12) and (1.10) we get the solution of this equation: $D^a = g_0 L^a$ with g_0 an arbitrary constant.

Therefore transformation (1.5) can be regarded as a group transformation on one hand and as generalized phase transformation (1.1) generated by the adjoint representation of the Lie group G on the other hand. Two different interpretations of transformation (1.5) are two ways of generalizing the work of Yang and Mills — two possibilities of theoretical constructions essentially different in its scope.

It should be also noted that a choice of generalized phase transformation (1.1) and a choice of some transformation of the group G are independent of each other so that one of these choices does not predetermine another one. In particular, if a field is subjected to a gauge transformation generated by some representation T of the Lorentz group, then the space-time coordinates do not change.

It is natural desire to have, for the quantities ψ_a from (1.1), not only a mathematical characteristics but as well physical or some other interpretation. Thus, regarding gauge transformations (1.1) that describe the invariance of the theory as a realization of the freedom inherent in the field $\Psi(x)$, I am likely to regard the quantities θ_a as encoding the ghost-carrier of freedom.

1.3 Basic assumptions about the theory of hadrons

Let us turn to hadrons and formulate our

First assumption. The theory of hadrons is a Lorentz-vector gauge theory, i.e. Lagrangian and equations that describe hadrons and their interactions are invariant under gauge transformations (1.1) generated by the vector representation of the full Lorentz group.

Note at once that there are two vector representations of the full Lorentz group: with polar and axial vectors. We will analyze both situations.

Let us now discuss the motives that lead me to this very assumption.

No group of internal symmetry can give a selection mechanism for some set of representations of the Lorentz group from continuum of available representations. Note that this problem is one of the important problems for us. Of course, an analysis of problems connected with internal symmetry is necessary. I believe however that these problems should be of secondary importance.

Why have we taken, for $\{\theta_a\}$ from (1.1), the vector representation of the full Lorentz group? Here I have the following three arguments.

First, I tried to look, from different points of view, at the most irreproachable gauge Yang-Mills theory — the theory of electroweak interaction with the group $SU(2) \otimes U(1)$. We mentioned above that gauge Yang-Mills transformations can be treated as transformation (1.1) generated by the adjoint representation of the group. If, instead of the group $SU(2) \otimes U(1)$, we take the group $SO(3) \otimes U(1)$ isomorphic to it, then the adjoint representation of the latter will be, at the same time, the fundamental representation. For the Lorentz group, the vector representation is the fundamental representation.

Second, any finite-dimensional irreducible representation of the Lorentz group with integer spins can be obtained from the direct product of vector representations. Therefore, restrictions following from the condition of Lorentz-vector gauge invariance will be not softer than restrictions appearing in the case of invariance under transformations (1.1) generated by any representation of the Lorentz group which is decomposable into a direct sum of finite-dimensional irreducible representations with integer spins.

The third argument is that it is very tempting to build a theory invariant under gauge transformations (1.1) generated by a first rank bispinor of the full Lorentz group (Lorentz-spinor gauge theory). Such a theory could realize the initial idea of supersymmetric approaches: interdependence of fermionic and bosonic fields, but without introduction of any spinor coordinates. This perspective cannot be ignored. A detailed analysis of such theories will be given in Section 3. Our conclusions are as follows: Lorentz-spinor gauge theories without interaction exist; Lagrangian invariant under Lorentz-spinor gauge transformations cannot involve three-particle interaction of baryons and mesons. The last conclusion forced me to give up the following consideration of these theories.

The requirement that a relativistically-invariant equation of the type

$$(\Gamma^\mu \frac{\partial}{\partial x^\mu} + \Phi) \Psi(x) = 0, \quad (1.13)$$

with Φ some structure that describes interaction with the field $\Psi(x)$, is invariant under Lorentz-vector gauge transformations (1.1) happens to be rather restrictive.

The system of algebraic equations following from this requirement is overdetermined so that from the very beginning it is impossible to say whether there is at least one its solution. It is extremely important that the representation S of the Lorentz group according to which the field $\Psi(x)$ transforms cannot be chosen in an arbitrary way. Admissible representations are found from the condition of solvability of the above mentioned system of algebraic equations. Note that each of solutions that are obtained for the matrix operators D^μ from (1.1) and for Γ^μ from (1.13), apart from the normalization factor, does not include arbitrary quantities. Section 2 of the present work is devoted to an analysis of the Lorentz-vector gauge invariance.

An essential ingredient of gauge theories is the Higgs mechanism by which the fields acquire masses. The first step that is tempting to be considered is to introduce the Lorentz-scalar Higgs boson ϕ_0 (whose Lorentz-vector gauge transformation is trivial: $D^\mu = 0$ in (1.1)) that has a non-zero vacuum expectation value λ_0 and interacts with the hadron field Ψ

$$\mathcal{L}_{int} = \bar{\Psi} \Gamma \Psi \phi_0. \quad (1.14)$$

Then the corresponding Gel'fand-Yaglom equation for the free field Ψ (in (1.13) it is necessary to put $\Phi = ig_0 \lambda_0 E$, with E the unit operator) gives, for each value of spin, a continuous mass spectrum $M \in (-\infty, -M_0) \cup (-M_0, +\infty)$, and M_0 is independent of spin that means an infinite degeneration (details are presented in Section 4 of the paper).

If the Higgs field is non-trivial in respect to Lorentz-vector gauge transformation (1.1), then, in accordance with the results of Section 2 of the work, this field should be infinite-dimensional. The formulated arguments stimulate me to formulate the following

Second assumption. There exists a non-trivial Higgs field which coincides with the field of hadron mesons.

In other words, it is supposed that Lorentz-scalar components of meson field have non-zero vacuum expectation values.

I restrict myself to considering the three-particle interaction of hadron fields

$$\mathcal{L}_{int\ hadr} = \bar{\psi} R^\mu \psi \phi_\mu + \bar{\phi} Q^\mu \phi \phi_\mu. \quad (1.15)$$

Let us here fix notations that we use: Ψ can be both fermion and meson field, ψ is a fermion field, and ϕ is a meson field.

Section 4 of the work is devoted to the problem of calculating of operators R^μ and Q^μ , starting from the condition of invariance of Lagrangian (1.15) under Lorentz-vector gauge transformations.

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НАЧАЛА КАЛИБРОВОЧНОЙ ТЕОРИИ РЕАЛЬНЫХ АДРОНОВ
I. Не-янг-миллсовы калибровочные теории. Основные
предположения относительно теории адронов

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